

Express + MySQL Deployment on Railway

Overview

This document explains **step-by-step** how to deploy a **MySQL database first** and then an **Express (Node.js) backend via GitHub** on **Railway**.

Architecture

Client (Postman / Frontend) → Express Backend (Railway) → MySQL Database (Railway)

STEP 1: Create Railway Project

1. Go to <https://railway.app>
 2. Login using GitHub
 3. Click **New Project**
 4. Choose **Empty Project**
-

STEP 2: Deploy MySQL Database (FIRST)

Add MySQL Service

1. Click **Add Service**
2. Select **Database** → **MySQL**
3. Wait for provisioning

You will now see:

- MySQL service (Online)
-

MySQL Environment Variables

Railway auto-generates variables like:

- MYSQLHOST
- MYSQLUSER
- MYSQLPASSWORD
- MYSQLDATABASE
- MYSQLPORT

⚠ Do NOT edit these.

STEP 3: Create Database Tables

Open: MySQL Service → Database → Connect → MySQL CLI

Run:

```
CREATE TABLE users (  
  id INT PRIMARY KEY AUTO_INCREMENT,  
  name VARCHAR(100) NOT NULL,  
  email VARCHAR(100) UNIQUE NOT NULL,  
  password VARCHAR(255) NOT NULL,  
  mobile VARCHAR(20),  
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
);
```

Verify:

```
DESCRIBE users;
```

STEP 4: Prepare Express Backend Locally

Install dependencies

```
npm install express mysql2 jsonwebtoken crypto-js cors
```

db/pool.js

```
const mysql = require("mysql2/promise");  
  
const pool = mysql.createPool({  
  host: process.env.DB_HOST,  
  user: process.env.DB_USER,  
  password: process.env.DB_PASSWORD,  
  database: process.env.DB_NAME,  
  port: process.env.DB_PORT,  
});  
  
module.exports = pool;
```

server.js

```
const express = require("express");
const cors = require("cors");

const userRouter = require("./routes/user");
const auth = require("./utils/auth");

const app = express();

app.use(cors());
app.use(express.json());
app.use(auth.authUser);

app.use("/users", userRouter);

const PORT = process.env.PORT || 3000;
app.listen(PORT, () => console.log("Server running"));
```

STEP 5: Push Express Backend to GitHub

```
git init
git add .
git commit -m "Express backend ready"
git branch -M main
git remote add origin <repo-url>
git push -u origin main
```

STEP 6: Deploy Express via GitHub on Railway

1. Railway → Add Service
2. Choose **GitHub Repo**
3. Select backend repository
4. Deployment starts automatically

STEP 7: Add DB Variables to Express Service

Add these variables in **Express** → **Variables**:

```
DB_HOST=mysql.railway.internal
DB_USER=<MYSQLUSER>
DB_PASSWORD=<MYSQLPASSWORD>
DB_NAME=<MYSQLDATABASE>
DB_PORT=<MYSQLPORT>
```

MySQL service var	Express var
MYSQL_HOST	DB_HOST
MYSQL_USER	DB_USER
MYSQL_PASSWORD	DB_PASSWORD
MYSQL_DATABASE	DB_NAME
MYSQL_PORT	DB_PORT

⚠ Use VALUES, not variable names.

STEP 8: Redeploy Express Service

Redeploy is mandatory after adding env variables.

STEP 9: Test APIs

Health

GET /users/health

Signup

POST /users/signup

```
{
  "name": "User",
  "email": "user@gmail.com",
  "password": "user123",
  "mobile": "9999999999"
}
```